

REQUIREMENT TRACEABILITY MATRIX (RTM)

Student Performance Analytics & Decision Support System

Purpose: The RTM ensures that every business requirement is mapped to functional requirements, user stories, acceptance criteria, and testing activities.

Requirement ID	Business Requirement	Functional Requirement	User Story	Acceptance Criteria	Test Case ID
BR001	Centralized student data	FR001 Student Management	US001	Student records stored successfully	TC001
BR002	Consolidate assessment data	FR003 Test Management	US004	Assessment data imported successfully	TC002
BR003	Automated score calculation	FR006 Score Calculation	US004	Calculated score matches expected result	TC003
BR004	Track student progress	FR007 Progress Tracking	US002	Performance trend displayed correctly	TC004
BR005	Identify weak students	FR008 Risk Identification	US006	Risk categories generated correctly	TC005
BR006	Subject analysis	FR009 Analytics Module	US003	Subject performance available	TC006
BR007	Chapter analysis	FR010 Chapter Analytics	US003	Chapter accuracy calculated correctly	TC007
BR008	Management dashboards	FR011 Executive Dashboard	US005	Dashboard displays KPIs	TC008

Requirement Coverage Summary

Category	Count
Business Requirements	8
Functional Requirements	12
User Stories	6
Acceptance Criteria	8
Test Cases	8

Requirement Status Tracking

Requirement ID	Priority	Status
BR001	High	Completed
BR002	High	Completed
BR003	High	Completed
BR004	High	Completed
BR005	High	Completed
BR006	Medium	Completed
BR007	Medium	Completed
BR008	High	Completed